Tutorial on

Programming Novel AI Accelerators for Scientific Computing

SC26
November 16, 2026

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Varuni K Sastry (ANL), Sylvia Howland (Cerebras), Tim Clarke (SambaNova), Moritz Thüning (Tenstorrent)
Taejoon Kong (Rebellions), Satyam Srivastava and Saliya Ekanayake (d-Matrix)

Argonne NATIONAL LABORATORY

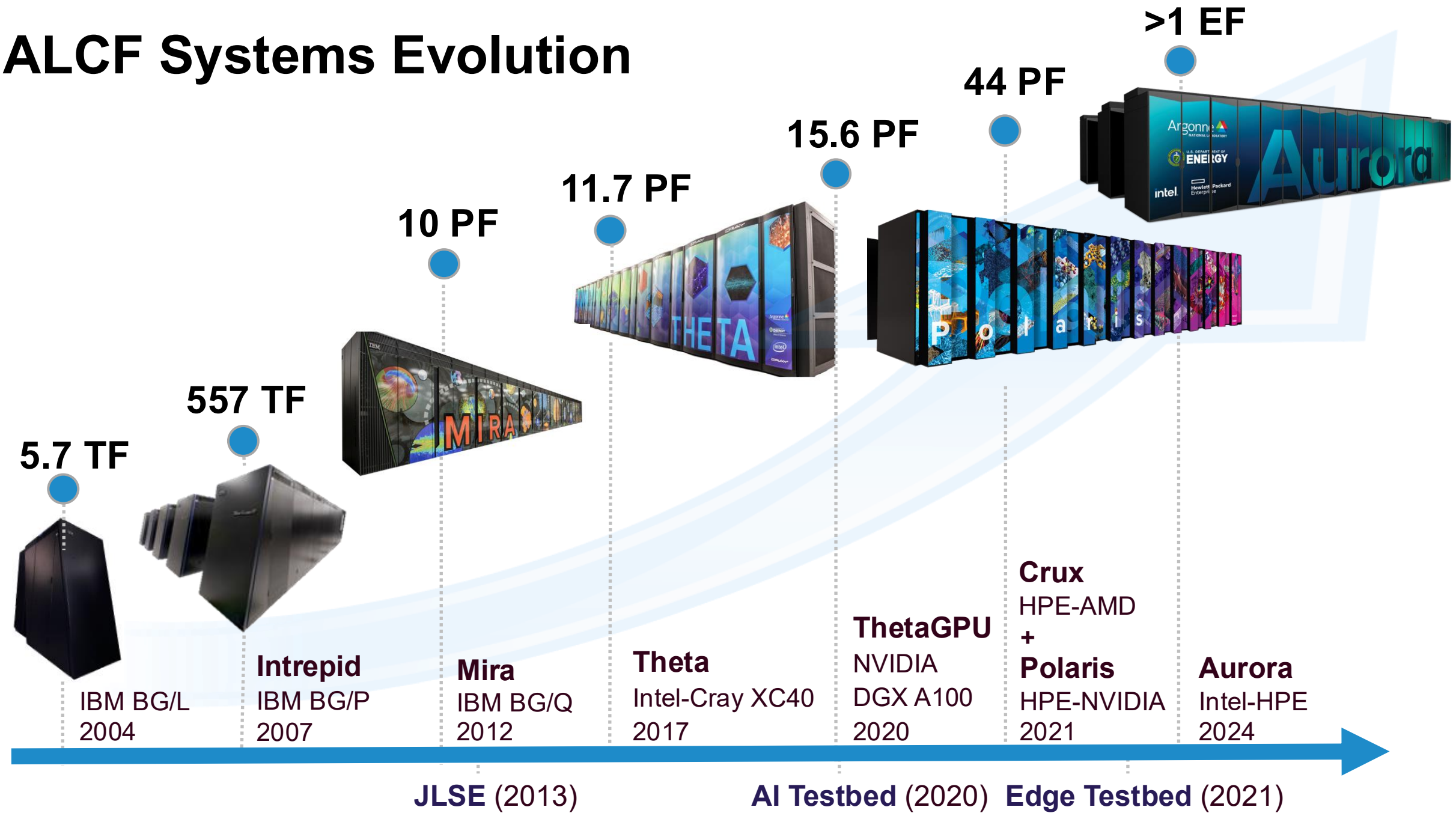
Tutorial Agenda

https://github.com/argonne-lcf/ALCF-AIAccelerators-tutorial

Agenda

Time	Topic	Speaker
8:30 AM - 8:45 AM	Welcome and Overview of the ALCF AI Testbed	Murali
8:45 AM - 10:00 AM	Cerebras	Sylvia
10:00 AM - 10:30 PM	-- Coffee Break --	
10:30 AM - 11:15 AM	SambaNova	Tim
11:15 AM - 12:00 PM	Discussion and Hands-on	Varuni, Murali
12:00 PM - 01:30 PM	-- Lunch Break --	
01:30 PM - 02:15 PM	Tenstorrent	Moritz
02:15 PM - 03:00 PM	Rebellions	Taejoon
03:00 PM - 03:30 PM	-- Coffee Break --	
03:30 PM - 04:15 PM	d-Matrix	Satyam, Saliya
04:15 PM - 05:00 PM	Discussion and Hands-on	Varuni, Murali

ALCF Systems Evolution



ALCF AI Testbed

https://www.alcf.anl.gov/alcf-ai-testbed



Cerebras



SambaNova



Groq



Graphcore



Tenstorrent

- Infrastructure of next-generation machines with hardware accelerators customized for artificial intelligence (AI) applications.
- Provide a platform to evaluate usability and performance of machine learning based HPC applications running on these accelerators.
- The goal is to better understand how to integrate AI accelerators with ALCF's existing and upcoming supercomputers to accelerate science insights

ALCF AI Testbed

ALCF AI Testbed Systems are in production and available for allocations to the research community

Training

- Cerebras



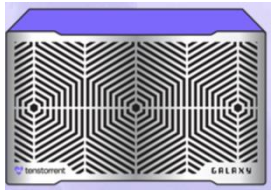
Cerebras CS-3
4 WSEs (wafer scale engine)

Inference

- SambaNova
- Groq
- Cerebras
- Tenstorrent



6 nodes
16 SN40L RDUs/node



2 Tenstorrent Galaxy
32 wormhole chips



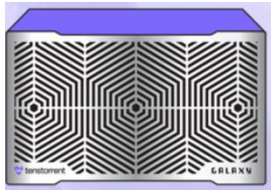
Cerebras CS-3 – 4 WSE
(upcoming support)

Low-level SDK

- Cerebras
- Tenstorrent



Cerebras CS-3



Tenstorrent Galaxy

Getting Started on ALCF AI Testbed

Available for Allocations

- Cerebras CS-3
- Sambanova Inference – Metis SN40L (Available with ALCF Inference Service Endpoints)
- Tenstorrent (coming soon)
- GroqRack
- Graphcore Bow Pod64

AI Testbed User Guide

Director's Discretionary (DD) awards

- Scaling code
- Preparing for future computing competition
- Scientific computing in support of strategic partnerships.

Allocation Request Form

https://www.alcf.anl.gov/science/directors-discretionary-allocation-program

NAIRR Pilot

Aims to connect U.S. researchers and educators to computational, data, and training resources needed to advance AI research and research that employs AI.

https://nairrpilot.org/

https://docs.alcf.anl.gov/ai-testbed/getting-started/

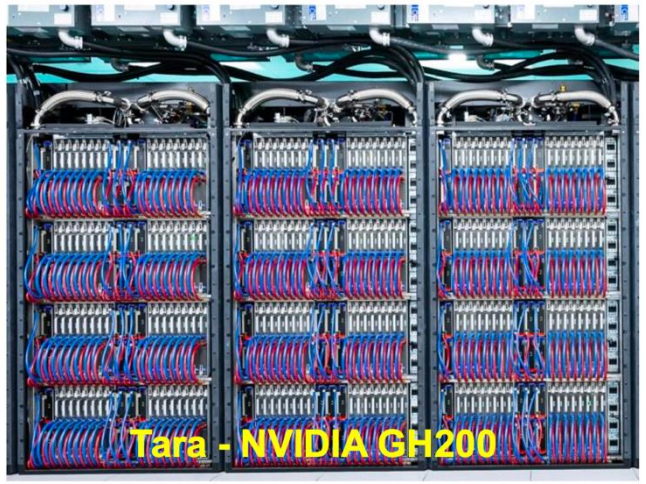
ALCF AI Testbed



The ALCF AI Testbed houses some of the most advanced AI accelerators for scientific research.
The goal of the testbed is to enable explorations into next-generation machine learning applications and workloads, enabling the ALCF and its user community to help define the role of AI accelerators in scientific computing and how to best integrate such technologies with supercomputing resources.
The AI accelerators complement the ALCF's current and next-generation supercomputers to provide a state-of-the-art computing environment that supports pioneering research at the intersection of AI, big data, and high performance computing (HPC).
The platforms are equipped with architectural features that support AI and data-centric workloads, making them well suited for research tasks involving the growing deluge of scientific data produced by powerful tools, such as supercomputers, light sources, telescopes, particle accelerators, and sensors. In addition, the testbed will allow researchers to explore novel workflows that combine AI methods with simulation and experimental science to accelerate the pace of discovery.

ALCF IS DEPLOYING DIVERSE INFERENCE SYSTEMS FOR SCIENCE









https://docs.alcf.anl.gov/services/inference-endpoints/

Useful Links

ALCF AI Testbed

- Overview: https://www.alcf.anl.gov/alcf-ai-testbed
- Guide: https://docs.alcf.anl.gov/ai-testbed/getting-started/
- Training:
  - https://www.alcf.anl.gov/ai-testbed-training-workshops
- Allocation Request: Allocation Request Form
- Support: support@alcf.anl.gov

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